

Predictive Locomotion Mode Recognition and Accurate Gait Phase Estimation for Hip Exoskeleton on Various Terrains

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Abstract—In recent years, lower-limb exoskeletons have been applied to assist people with weak mobility in daily life, which requires enhanced adaptability to complex environments. To achieve a smooth transition between different assistive strategies and provide proper assistance at the desired timing during locomotion on various terrains, two significant issues should be addressed: the delay of locomotion mode recognition (LMR) and the accuracy of gait phase estimation (GPE), which are yet critical challenges for exoskeleton controls. To tackle these challenges, a high-level exoskeleton control, including a depth sensor-based LMR method and an adaptive oscillator-based GPE approach, is developed in this study for terrain-adaptive assistive walking. An experimental study was conducted to evaluate the effectiveness and usability of the proposed control in a real-world scenario. Experimental results suggested that the depth sensor-based LMR method can detect the locomotion mode change 0.5 step ahead of the assistive strategy switch of the leading leg, while the average environment classification accuracy across five subjects was higher than 98%. The accuracy is comparable with the state-of-the-art LMR methods, but its predictive capability is beyond existing LMR methods applied in lower-limb exoskeletons. Moreover, the adaptive oscillator-based GPE approach accurately estimated the user's gait phase during

locomotion on various terrains, with a root-mean-square (RMS) gait phase reset error of only $\pm 0.27\%$, outperforming the literature standard.

Index Terms—Gait phase estimation (GPE), hip exoskeleton, locomotion mode recognition (LMR), various terrains.

I. INTRODUCTION

OVER the past decades, lower-limb exoskeletons have been regarded as a promising technology to assist human mobility. With latest technological advancements, lower-limb exoskeletons are no longer confined to experimental environment or fixed application scenario, such as treadmill-based gait training. More and more lower-limb exoskeletons have been applied to assist people with weak mobility in daily life, which requires enhanced adaptability to complex environments [1].

The main role of a powered lower-limb exoskeleton is to provide proper assistance (in terms of magnitude and timing) to the user's joints, and existing studies [2], [3] have demonstrated that both the force magnitude and timing are crucial for maximizing user benefit. When navigating on various terrains (e.g., level ground, stairs, and ramps), each locomotion mode corresponds to different joint kinetic and kinematic patterns [4], and usually requires different assistive strategies [5], [6]. In order to be effective in daily-life activities and decrease the risk of accidental injury, the exoskeleton controller should be able to timely detect the locomotion mode of the user, and precisely estimate the gait phase adapting to changes caused by locomotion mode transition and intracycle variability [7].

To date, various methods from manual switching methods to automatic switching strategies have been developed for lower-limb exoskeletons that assist human locomotion on various terrains. Ideally, transition between different locomotion modes should be automatic, continuous, and seamless, while suitable assistance is provided. But this is challenging to be achieved due to the presence of locomotion mode detection delay. Most of developed methods, such as threshold-based or rule-based methods, recognize locomotion modes based on the human states (e.g., joint angles, heel contact, etc) measured by mechanical sensors, which can cause up to one step delay in locomotion mode recognition (LMR) [8]. Currently, the state-of-art rule-based LMR approaches (in terms of the detection latency) for lower-limb exoskeleton or prosthesis were introduced in [9],

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[10]. Although the locomotion mode can be fast identified at the beginning of a new step, the detection latency was still as high as $9.06 \pm 3.46\%$ [9], $32 \pm 8.3\%$ [10] of a step. In contrast to the LMR methods utilizing signals from mechanical sensors, surface electromyography (sEMG)-based LMR methods possess a shorter detection latency since the sEMG appears prior to the mechanical motion due to the response time of the musculoskeletal system. However, the reliability of sEMG sensing is significantly affected by skin condition, humidity, electrode location, etc. [10], [11], and is yet a critical challenge for sEMG-based LMR methods.

In order to address the aforementioned limitations of existing LMR methods, a depth sensor-based LMR method is developed in this study, which possesses predictive capability during transitions between different locomotion modes. The term ‘predictive’ means that the new locomotion mode can be detected before the transition starts. Although existing studies have adopted a depth version system to assist the control of powered prosthesis [12] or help lower-limb exoskeleton crossing the obstacle [1], to our knowledge, this work is the first to introduce a depth sensor-based LMR method into the control of lower-limb exoskeletons and evaluate its predictive capability during locomotion mode transitions through experiments.

Accurate and predictive LMR enables the exoskeleton controller to switch the assistive strategy seamlessly during locomotion mode transitions. Nevertheless, the effectiveness of exoskeleton assistance also highly relies on the accuracy of gait phase estimation, especially during transition gait cycles. Different approaches have been adopted to estimate the user’s gait phase in the literature, of which time-based estimation (TBE) method is the most widely used method. For the TBE method, mechanical sensors, such as force-sensitive insole or inertial measurement unit (IMU), are utilized to detect characteristic gait event as the beginning of a new step. Average stride time can be calculated from several previous steps, and the user’s gait phase is estimated from comparison of the time since last characteristic gait event and the average stride time. This method is easy to be implemented, but lacks ability of adjusting to walking speed changes quickly, since several steps need to be taken for the average stride time to converge to a new value.

In view of the limitation of the TBE method, a more complex gait phase estimation method using an adaptive oscillator (AO) has been investigated and adopted in the control of lower-limb exoskeletons [13]–[17]. The AO-based method learns the envelope and the frequency from the teaching signal, such as the user’s joint angle, and generates an estimation of the user’s gait phase. Existing studies have demonstrated that this is a robust method of adapting to walking speed changes [15], [18], [19]. However, few studies [13] have applied the AO-based approach to estimate the user’s gait phase when navigating on various terrains and quantitatively evaluated its performance, especially during locomotion mode transitions. Recently, some research groups have introduce machine learning technologies into the gait phase estimation during assistive walking on various terrains [20]. By utilizing meaningful information from a variety of sensors, the machine learning algorithm comprehends the user’s gait phase accurately. However, these methods usually



Fig. 1. Powered hip exoskeleton and sensing system.

require a large set of training data from the users to guarantee the high accuracy of gait phase estimation, which might not feasible for people with weak mobility and hinders their applications in practice [7], [10]. In this work, an AO-based method is developed for accurate gait phase estimation when navigating on various terrains. Different from most of existing AO-based methods, the angular velocity instead of the angle is adopted as the teaching signal to speed up the adaptation to gait pattern changes, resulting in more accurate gait phase estimation during locomotion on various terrains.

The main purpose of this study is to develop a high-level control of powered hip exoskeleton for assistive walking on various terrains, which can detect the user’s locomotion mode change in advance and provide accurate gait phase estimation during multi-mode locomotion. Experiments are performed to validate the effectiveness of the proposed control in a real-world scenario, and experiment results are discussed.

II. METHODS

A. Powered Hip Exoskeleton and Control Architecture

The powered hip exoskeleton (Ant-C1 Pro-E, Kenqing technology, China) and sensing system are shown in Fig. 1, which weighs 3.8 Kg totally. The actuators can provide a peak torque of up to 15 Nm for hip flexion and extension through the links extended down to the thigh, while the passive joints are added to provide two passive degree of freedom for hip abduction and adduction during walking. The sensors used for locomotion mode recognition and gait phase estimation include a RGBD camera (RealSense D435, Intel, USA) and three 6-axis IMUs (MTi 630, Xsens, Netherlands). Specifically, one IMU is equipped on the adjustable support to measure the spatial posture of RGBD camera, while the other two IMUs are mounted on thighs to measure the angle and angular velocity of hip joints. The real-time control computer generates IMU data via

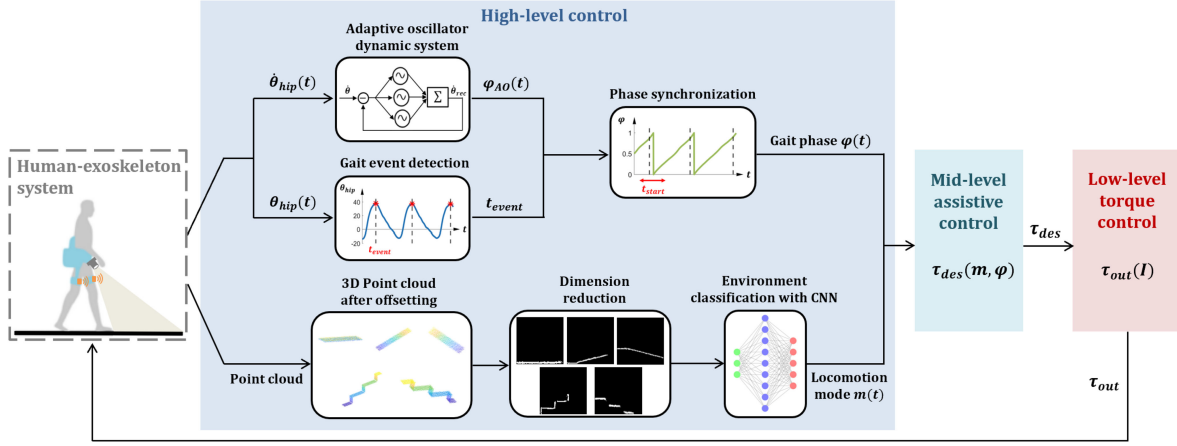


Fig. 2. Control architecture of the powered hip exoskeleton. The high-level control, mainly consisting of a depth sensor-based locomotion mode recognition module and an adaptive oscillator-based gait phase estimation module.

a controller area network (CAN) bus at a frequency of 400 Hz, and communicates with the RGBD camera via a USB-C cable to collect depth data at a frequency of up to 90 Hz. Then the raw data is processed accordingly and the generated control commands are sent to the Raspberry Pi, the main controller of the powered hip exoskeleton.

As illustrated in Fig. 2, the control architecture of the powered hip exoskeleton can be broken into three layers according to different function: High, mid, and low-level controls. The high-level control implements algorithm to estimate the user's intent and state, such as locomotion mode $m(t)$ and gait phase $\varphi(t)$. Then the mid-level control will generate torque control commands according the user's intent and state. Finally, the lower-level control receives the commanded torque and generates assistive torque accordingly through a closed-loop PID torque controller. This study focuses on the high-level control, mainly including a depth sensor-based locomotion mode recognition module and an adaptive oscillator-based gait phase estimation module.

B. Depth Sensor-Based Locomotion Mode Recognition

1) **3D Point Cloud Generation and Preprocessing:** The environmental information in front of the user, which provides environmental context for locomotion mode recognition, can be obtained by capturing three-dimensional (3D) environmental point cloud with the RGBD camera fixing on the user's waist. The visual field of the RGBD camera is limited to 0–1.0 m in front of the user based on the step length of adults. To reduce misclassification and computational load, the original 3D point cloud needs to be preprocessed as follows before applying to environmental recognition.

To decrease the change of point cloud caused by the user's motion and help distinguish level ground from ramp, the point cloud is offset from the camera coordinate system to the invariant ground coordinate system in real time according to the Euler angle measured by the IMU fixed together with the RGBD camera:

$$p^{Ground} = R_{Camera}^{Ground} \cdot p^{Camera} \quad (1)$$

where p^{Ground} and p^{Camera} are the environmental point cloud in the ground and camera coordinate system, respectively; R_{Camera}^{Ground} indicates the rotation matrix from the camera coordinate system to the ground coordinate system.

In addition, directly applying 3D point cloud for environment classification requires huge computational resource for its unstructured and disordered nature. Therefore, to convert the point cloud into structured and ordered data and reduce the computational effort, the 3D point cloud is projected onto a two-dimensional (2D) plane and converted to binary images. As described in [21], the 2D projection of the point cloud in the sagittal plane (x-o-z plane) can provide sufficient information for environment classification. The point cloud can be converted to a binary image of 100×100 pixels as:

$$img(r, c) = \begin{cases} 1, & 0.01(c-1) \leq x_i - x_{min} < 0.01c \text{ \& } 0.01(r-1) \leq z_i - z_{min} < 0.01r \\ & (i = 1, 2, \dots, n) \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where $img(r, c)$ represents the pixel in row r and column c of the binary image and each pixel of the image corresponds to 0.01 m in the real world; x_i and z_i are the x and z coordinates of the i -th point in the 2D point cloud composed of n points; x_{min} and z_{min} represent the minimum x and z coordinates of the point cloud, respectively.

2) **Environment Classification Based on CNN:** After converting 3D environmental point cloud to binary images, it is necessary to find a classification algorithm to accurately classify these images. In recent studies, deep learning algorithms such as convolutional neural networks (CNNs) have received increasing attention in various domains such as image classification [25] and semantic segmentation [26] for their high accuracy and ability to automatically extract deep features. In addition, the parameter sharing method of the convolutional layers helps CNN learn effectively.

Due to the attractive qualities of CNN, in this paper, a CNN is designed and trained to classify environmental binary images. As shown in Fig. 3, there are three convolutional layers in two different types in the designed CNN. The kernel sizes of two

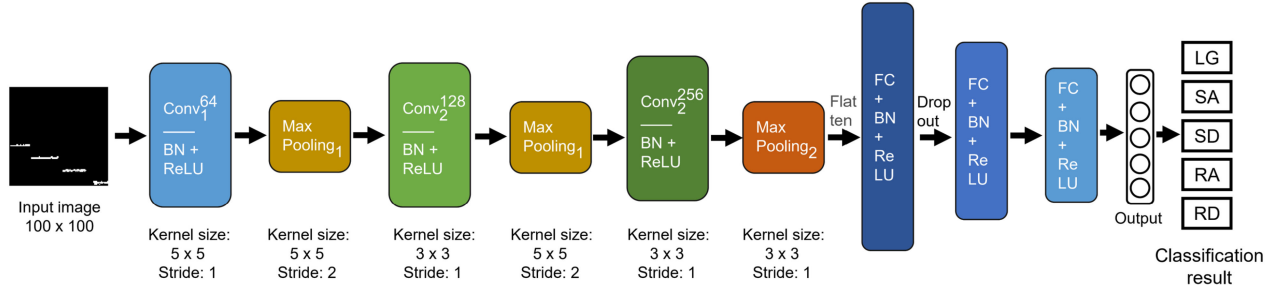


Fig. 3. The schematic of the proposed CNN. The superscript and the subscript of a convolutional layer indicates its type and parameter, e.g., Conv_1^{64} indicates a type-one convolutional layer with 64 channels. The batch normalization layer, rectified linear unit layer and fully connected layer is described as BN, ReLU and FC respectively.

types of convolutional layers (Conv_1 and Conv_2) are 5×5 pixels and 3×3 pixels, respectively, and the strides of all convolutional layers are 1. Each convolutional layer is followed with a batch normalization (BN) layer for enhancing generalization ability, a rectified linear unit (ReLU) and a max-pooling layer for downsampling. Two types of max-pooling layer (MaxPooling_1 and MaxPooling_2) consist of a kernel size of 5 and a stride of 2, and a kernel size of 3 and a stride of 1, respectively.

The hidden features from the last max-pooling layer are flattened and then fed into 3 fully connected (FC) layers and an output layer, which reduces the feature dimension and finally maps the features to classification scores of 5 different environment classes. The environment class with the highest score is the classification result. BN and ReLU are applied after each FC layer and a dropout layer is set after the first FC layer to avoid over-fitting. In order to reduce misclassification caused by incidental factors such as obstacles and rotation offset mistakes, a majority filter is implemented to smooth the classification results and further enhance the accuracy. The filter smooths the results by sliding a window and replacing the result of each frame with the mode of all classification results in the window. In the present paper, the length of the window is taken as 6. Finally, the filtered classification result is adopted as the predicted locomotion mode $m(t)$, which is utilized to switch the assistive strategy predefined for each specific locomotion mode (LG - Level Ground, SA - Stair Ascent, SD - Stair Descent, RA - Ramp Ascent, RD - Ramp Descent) at the beginning of swing phase during a transition gait cycle.

C. Adaptive Oscillator-Based Gait Phase Estimation

As introduced in Section II-A, the powered hip exoskeleton employs two thigh-mounted IMUs to measure the angle and angular velocity of hip joints in the sagittal plane. Then the user's gait phase is estimated by the proposed AO-based method, which consists of three main parts: Adaptive oscillator dynamic system, gait event detection, and phase synchronization (see Fig. 2).

1) *Adaptive Oscillator Dynamic System*: In this work, the measured hip angular velocity is utilized as the teaching signal of the AO dynamic system to extract continuous phase of gait cycle in real time. The AO dynamic system is an error-driven

system and thus the input is

$$e(t) = \dot{\theta}_{hip}(t) - \hat{\theta}_{hip}(t) \quad (3)$$

where $\dot{\theta}_{hip}(t)$ is the angular velocity measured by IMU and $\hat{\theta}_{hip}(t)$ is the reconstructed angular velocity given by a finite-term Fourier decomposition performed in real time.

Specifically, signal characteristics and the reconstructed teaching signal are obtained using the following equations:

$$\dot{\phi} = \omega - \psi e(t) \cos(\phi) \quad (4)$$

$$\dot{\omega} = -\psi e(t) \sin(\phi) \quad (5)$$

$$\dot{\alpha}_k = \epsilon \cos(k\phi) e(t) \quad (k = 0, \dots, N_f) \quad (6)$$

$$\dot{\beta}_k = \epsilon \sin(k\phi) e(t) \quad (k = 0, \dots, N_f) \quad (7)$$

$$\hat{\theta}_{hip} = \sum_{k=0}^{N_f} \alpha_k \cos(k\phi) + \beta_k \sin(k\phi) \quad (8)$$

where ϕ is the oscillator phase, ω is the intrinsic frequency of the oscillator, and ψ , ϵ are gains determining the learning speed of the phase and frequency, respectively. The Fourier decomposition consists of N_f pairs of coefficients α_k and β_k , which are updated online using (6) and (8).

To obtain a gait phase ϕ_{ao} varying from 0 to 2π along a stride, the phase from AOs is normalized in the interval $[0, 2\pi)$:

$$\phi_{ao}(t) = \text{mod}(\phi(t), 2\pi). \quad (9)$$

2) *Gait Event Detection*: This part is responsible for identifying the beginning of each gait cycle. The characteristic gait event that most commonly utilized as the beginning of a new gait cycle is heel strike (HS). But it is challenging to detect HS accurately and robustly using only body-mounted sensors above the knee joint [22]. Consequently, the maximum hip flexion/extension (MHF/MHE) are widely adopted as the characteristic gait event for hip exoskeletons. In this work, MHF is detected in real time and used for gait cycle segmentation.

3) *Phase Synchronization*: Ideally, the gait phase ϕ_{ao} should linearly increases from 0 to 2π , where 0 corresponds to the detected gait event. But mismatch between the gait phase estimated by AOs and the detected gait event is usually unavoidable in practice. When the characteristic gait event is detected, the

mismatch $P_e(t_{event})$ can be calculated as follows:

$$P_e(t_{event}) = \begin{cases} -\phi_{ao}(t_{event}), & \text{if } 0 \leq \phi_{ao} < \pi \\ 2\pi - \phi_{ao}(t_{event}), & \text{if } \pi \leq \phi_{ao} < 2\pi \end{cases} \quad (10)$$

Then the mismatch $P_e(t_{event})$ is corrected using the following equations for the synchronization of the phase estimation to the user's gait [18]:

$$\dot{\delta}_\phi = K_\phi \cdot [P_e(t_{event}) - \delta_\phi] \cdot e^{-\omega(t)(t-t_{event})} \quad (11)$$

$$\phi_{crt}(t) = \phi_{ao}(t) - \delta_\phi(t) \quad (12)$$

in which K_ϕ is a proportional gain, t_{event} is the timing when last characteristic gait event is detected, ω is intrinsic frequency of the AO, and ϕ_{crt} is the corrected gait phase estimation. After the phase synchronization, the starting point of each gait cycle will be locked to the characteristic gait event.

For convenience, the final gait phase is normalized into $[0,1)$ and noted as $\varphi(t)$:

$$\varphi(t) = \frac{\text{mod}(\phi_{crt}(t), 2\pi)}{2\pi} \quad (13)$$

Then combined with the identified locomotion mode $m(t)$, the terrain-adaptive assistive profile can be expressed as functions of $\varphi(t)$ and $m(t)$ as shown in Fig. 2.

III. EXPERIMENTS AND RESULTS

A. Experimental Protocol

To validate the effectiveness of the proposed control in locomotion mode recognition and gait phase estimation during locomotion on various terrains, an experimental study was conducted. The experimental protocol was approved by the Institutional Review Board at Southern University of Science and Technology, and informed consent was obtained from all subjects before completing the experimental protocol.

For experimental validation of the depth sensor-based LMR method, one healthy male subject (subject 0; age 25 years; weight 65 kg; height 170 cm) was recruited to collect the dataset used for the training and preliminary validation of the proposed CNN. In order to demonstrate the usability of the proposed depth sensor-based LMR method in a real-world scenario, both indoor and outdoor experiments were performed to collect the dataset (see Fig. 4). Finally, the dataset consisted of 8,500 binary images of indoor and outdoor environments, with 1,700 images of each type of daily-life terrains. Before training, the dataset was stochastically divided into a training set (80%) and a validation set (20%). The validation set is used for preliminary measurement of the classification performance and optimization of the hyper-parameters. Specifically, training options and hyper-parameters are described as follows. The Adam algorithm is used as the optimizer with a learning rate of 0.0002 and a weight decay of 0.0005. The maximum epoch for training is 50 and the batch size is set as 64. The proposed CNN is designed and trained on Python 3.7, CUDA 10.1 and PyTorch 1.8.0. The training process and the experimental analysis are carried out on a laptop computer equipped with an AMD Ryzen 7 4800H CPU (2.9 GHz), 16 GB RAM and an NVIDIA Geforce RTX 2060 GPU.

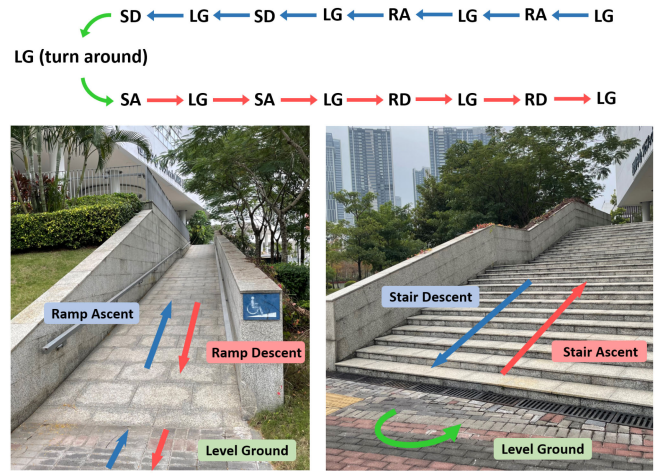


Fig. 4. Outdoor experimental environment and various terrains the subject experienced during one trial.

For final experimental validation of the proposed depth sensor-based LMR method, another five healthy male subjects (subject 1–5; age 23.6 ± 1.3 years; weight 63.0 ± 9.7 kg; height 169 ± 5 cm; and mean $\pm$ SD) were recruited to walk in the same indoor and outdoor environment, and the terrain in front of the subjects was classified with the trained CNN directly to demonstrate the user-independence of the proposed depth sensor-based LMR method.

For experimental validation of the AO-based gait phase estimation method, only IMU data collected from the outdoor environment was analyzed because IMU signals were usually unaffected by the operating environment, and the angular velocity signal was filtered by a lowpass filter with a cutoff frequency of 1 Hz before inputted to the AO dynamic system. Each of the five subjects (subject 1–5) was asked to perform four trials at their preferred walking speed in the outdoor experimental environment, and each trial had 108 ± 8 steps on average. During one trial, the subjects walked through five types of daily-life terrains (LG, SA, SD, RA, and RD) and experienced eight types of locomotion mode transitions (LG to SA/SD/RA/RD and SA/SD/RA/RD to LG). Designing an assistive strategy and evaluating its effectiveness are beyond the purpose of this study, thus during all trials, the IMU data was recorded with the powered hip exoskeleton working in a zero impedance mode.

In previous literature studies, the TBE strategy was widely adopted to estimate user gait phase due to its ease of implementation. In view of this, the TBE method was also implemented in this study to provide baseline information for comparison purposes. To accelerate the convergence of average stride time and minimize the delay in responding to walking speed change during locomotion mode transitions, the TBE method used the stride duration from only one previous gait cycle. The root-mean-square (RMS) phase reset error across all strides was adopted as the evaluation metric, following the definition reported in [19]:

$$RMS_{GPE} = \sqrt{\frac{\sum_{i=1}^{N_s} (P_e(i) - P_r)^2}{N_s}} \quad (14)$$

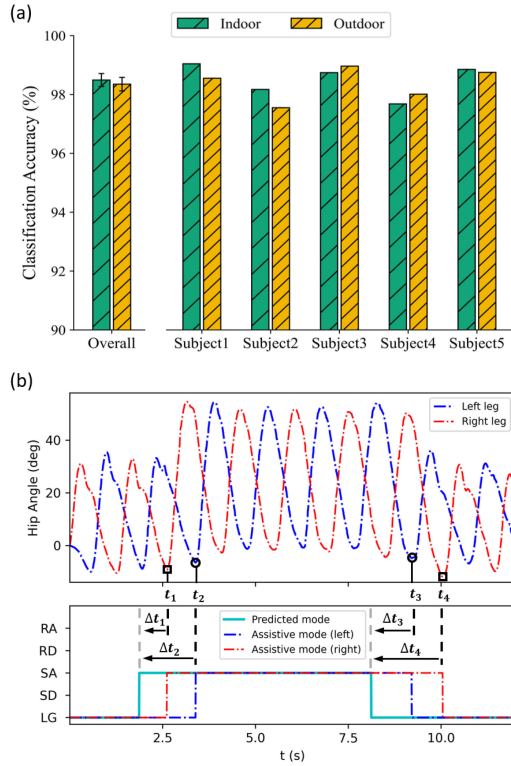


Fig. 5. Results of locomotion mode recognition. (a) Environment classification results of the indoor and outdoor experiments of all subjects (subject 1–5). Error bars of overall results represent $\pm$ standard error of the mean (SEM) across all subjects. (b) Results of the depth sensor-based locomotion mode recognition from one subject during walking on LG and SA. Specifically, the right leg is the leading leg during the LG to SA transition while the left leg is the leading leg during the SA to LG transition. t_i ($i = 1, 2, 3, 4$) represents the timing of maximum hip extension (corresponding to toe-off as the beginning of swing phase) during transition gait cycles. Δt_i represents the leading time of locomotion mode change detection.

where RMS_{GPE} represented the root-mean-square phase reset error across N_s strides; $P_e(i)$ was the phase reset error of the i – th stride calculated according to (10); P_r was the reference/ideal gait phase at the detected gait event, which was 0 rd in this study because the detected gait event was regarded as the starting of a new gait cycle. The accuracy of gait phase estimation was evaluated within each steady-state locomotion mode and during a transition gait cycle between different locomotion modes.

B. Results of Locomotion Mode Recognition

1) *Analysis of Environment Classification:* The environment classification performance of the proposed depth sensor-based LMR method was validated through the indoor and outdoor experiments. As shown in Fig. 5(a), average classification accuracy of all subjects was $98.49 \pm 0.22\%$, $98.35 \pm 0.23\%$ for the indoor and outdoor experiments, respectively. Comparing the classification results of the indoor and outdoor experiments, no significant change in the classification accuracy was observed, demonstrating the real-world usefulness of our approach. During the experiments, the 3D cloud point generation and preprocessing took about 20 ms and the classification of each binary image could be finished within 3 ms.

2) *Analysis of Locomotion Mode Prediction:* The predictive capability of the proposed depth sensor-based LMR method during the locomotion mode transitions was also evaluated by experiments. As shown in Fig. 5(b), the locomotion mode transition can be predictively detected prior to the beginning of swing phase during the transition gait cycle so that the mode-dependent controller can switch the assistive mode for the following swing phase. The leading time (Δt_i in Fig. 5(b)) was defined as the interval between the timing when the locomotion mode change is detected and the timing the assistive mode switch occurred. Specifically, during the LG to RA transition, the leading time was $\Delta t_1 = 0.74$ s = 0.51 gait cycle (GC) and $\Delta t_2 = 1.51$ s = 0.97 GC for the leading leg and the following leg, respectively. During the RA to LG transition, the leading time was $\Delta t_3 = 1.09$ s = 0.76 GC and $\Delta t_4 = 1.92$ s = 1.34 GC for the leading leg and the following leg, respectively. Similar leading time was also observed during other types of locomotion mode transitions, and the locomotion mode change could be detected 0.5 GC ahead of the assistive mode switch of the leading leg on average, clearly demonstrating the predictive capability of the proposed depth sensor-based LMR method.

C. Results of Gait Phase Estimation

1) *Analysis of Continuous Walking on Various Terrains:* During continuous walking on various terrains, the TBE method had an average RMS_{GPE} across all subjects of $6.76 \pm 0.57\%$. In contrast, the proposed AO-based estimation method had an average RMS_{GPE} across all subjects of $4.12 \pm 0.27\%$, which is $38.99 \pm 4.03\%$ lower than the TBE method (Fig. 6(a)). For each specific locomotion mode, the proposed AO-based estimation method on average had a $69.97 \pm 1.70\%$, $20.40 \pm 6.92\%$, $47.64 \pm 7.49\%$, and $50.34 \pm 4.60\%$ lower RMS_{GPE} than the TBE method across all subjects, for SA, SD, RA, and RD, respectively (Fig. 6(b)). However, it is worth noting that the proposed AO-based estimation method on average had a $8.87 \pm 5.84\%$ higher RMS_{GPE} than the TBE method across all subjects for LG. This is not surprising because the TBE method usually yields high accuracy of gait phase estimation during steady-state walking such as LG walking. For the proposed AO-based estimation method, its adaptability to changes in walking speed and locomotion mode does not bring any benefit under such a condition.

2) *Analysis of Transition Gait Cycle:* During transition gait cycles, the proposed AO-based estimation method on average had a $64.68 \pm 4.41\%$, $34.53 \pm 17.37\%$, $35.85 \pm 10.22\%$, $39.76 \pm 12.81\%$, $12.25 \pm 12.36\%$, $17.54 \pm 9.69\%$, $4.03 \pm 19.89\%$, and $34.40 \pm 7.66\%$ lower RMS_{GPE} than the TBE method across all subjects, for LG to SA/SD/RA/RD and SA/SD/RA/RD to LG, respectively (Fig. 6(d)). In general, the proposed AO-based estimation method showed greater advantages during the LG to SA/SD/RA/RD transitions.

IV. DISCUSSION AND CONCLUSION

In this study, we aim to develop an effective control of powered hip exoskeleton to assist people with weak mobility in daily life. When navigating on various daily-life terrains, a

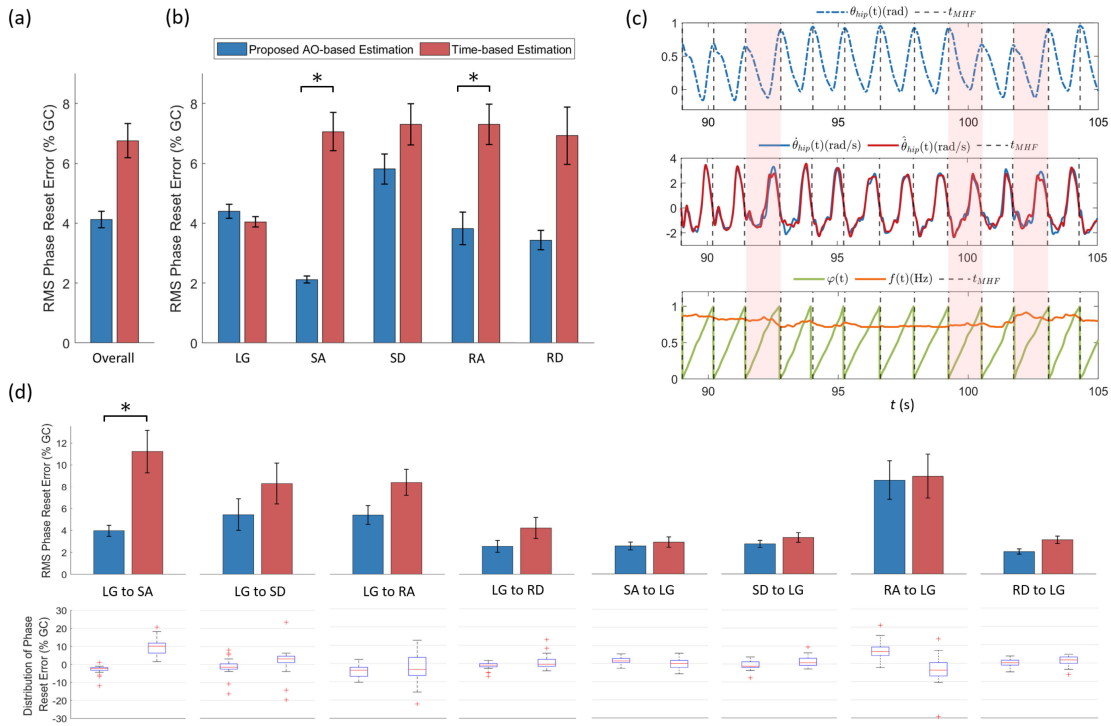


Fig. 6. Results of gait phase estimation. (a) Overall gait phase estimation performance for the proposed AO-based estimation method (blue) and the time-based estimation method (red). (b) Gait phase estimation performance for each locomotion mode. (c) Results of the proposed AO-based estimation from one subject during walking on LG and SA. Transition gait cycles (noted as T1, T2, and T3 in sequence) are highlighted. Specifically, T1 and T3 represent LG to SA transitions, and T2 represents SA to LG transition. (d) Gait phase estimation performance during transition gait cycles. Error bars represent ± 1 SEM, and asterisks indicate statistically significant difference ($p < 0.05$).

high-level control that is able to predictively detect the locomotion mode change and precisely estimate the user's gait phase during multi-mode locomotion plays a critical role in adapting gaits to varying terrains.

A. Advantages of the Proposed Method

Timely and accurate LMR has long been a goal of lower-limb exoskeleton research, with the majority of efforts focused on the rule-based or machine learning-based LMR methods utilizing human signals measured by mechanical or bioelectrical sensors. Unfortunately, the detection latency is yet a critical challenge and these LMR methods do not possess the ability to detect locomotion mode change in advance, which is crucial for the exoskeleton control during walking on varying terrains. To address this issue, a depth sensor-based LMR method is proposed in this paper. The biggest advantage of the proposed LMR method is its capability to predict locomotion mode change, completely eliminating the transitional delay as shown in Fig. 5(b). Another important advantage of the proposed depth sensor-based LMR method is its user-independence. In the experiments of environment classification, only the data collected from one additional subject (subject 0) was added to the training set, and then the trained CNN was directly applied to the environment classification for the other five subjects. However, for both the indoor and outdoor experiments, the average classification accuracy across the five subjects was still higher than 98%, which outperforms

existing depth sensor-based terrain detection methods [21], [23], [24], and is also comparable with the accuracy of the rule-based or machine learning-based LMR methods applied in lower-limb exoskeletons [8], [10], [25].

According to the results of locomotion mode prediction, the mode-dependent controller can seamlessly switch the assistance strategy for different locomotion mode. But the effectiveness of the phase-locked assistance highly relies on the accuracy of gait phase estimation. In order to adapt to changes caused by locomotion mode transition and intracycle variability, adaptive oscillator is adopted to estimate the user's gait phase in this work. As introduced in Section II-C, adaptive oscillator is an error-driven dynamic system. When the teaching signal changes its periodic behavior, the gait phase estimation is delayed until it converges to a new equilibrium. The convergence process can be shortened by simply increasing gains, which comes with trade-off in accuracy and robustness. For the proposed AO-based estimation method, the hip angular velocity instead of the hip angle is adopted as the teaching signal to drive the AO dynamic system, because the hip angular velocity is a periodic signal that changes less compared to the hip angle when navigating on varying terrains as shown in Fig. 6(c). This unique characteristic significantly speeds up the convergence process and reduces phase estimation delay during locomotion mode transitions. Consequently, not only our AO-based estimation method outperformed the TBE method within most of specific locomotion modes (except LG) by reducing the error rate by

47.1%, but also had better performance compared to the TBE method during transition gait cycles, reducing the error rate by 30.4%.

In previous literature studies, the most advanced gait phase estimation method using the non-user-specific AOs was presented in [15], [19], which has a gait phase estimation error rate of about 4%. However, this method was only evaluated during treadmill walking with different/changing speeds, which is still different from a real-world locomotion on various terrains. At present, the state-of-art methods in estimating gait phase during locomotion on various terrains is to use the machine learning technologies, which has a gait phase estimation error rate of 5% [20]. However, this method usually require a large set of training data from the users, which might not feasible for people with weak mobility and thus hinders its applications in practice [7], [10], [20]. In contrast, the AO-based estimation method presented in this work does not require any data collection before using. It has been tested in an outdoor environment including five daily-life terrains and eight different locomotion mode transitions, and yielded a gait phase estimation error rate of 4.12%, which validates its usability and effectiveness in a real-world scenario.

By combining the user-independent LMR method based on the depth sensor and the gait phase estimation method based on the non-user-specific AOs, the proposed exoskeleton control does not require any data collection or parameter tuning for a new exoskeleton user before using. Moreover, the proposed exoskeleton control only utilizes the sensors placed inside the device interface region (thigh-mounted IMUs, waist-mounted RGBD camera and IMU). In view of the aforementioned characteristics, we believe that our exoskeleton system is more convenient and promising from an perspective of real-world application.

B. Limitations and Future Works

One limitation of this study is that all experiments are performed with the powered hip exoskeleton working in a zero impedance mode. With exoskeleton assistance applied, the IMU data and the performance of gait event detection might be affected. Another limitation is that the algorithms for locomotion mode recognition and gait phase estimation are still deployed on a laptop computer for preliminary experimental evaluation. Further test of the algorithms on a wearable device, such as Raspberry Pi, is necessary to built a completely self-contained hip exoskeleton system. In the future, we will focus on designing a terrain-adaptive assistive strategy in the mid-level control and evaluating its performance together with the developed high-level control.

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